

The Great Bengal Famine and the Question of FAD Yet Again

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Abstract

The famine of 1943 cost the Bengal province some three millions lives. Following Amartya Sen's proposition that the famine was not caused by a significant decline in food availability as such the last two decades or so have witnessed a renewed scholarly interest in the subject. On the basis of a fresh look at the available crop statistics this essay supports his basic contention. But at the same time it argues that his proposition is not quite new. Secondly, it suggests that for a fuller understanding of the severity of the crisis it will not be enough to analyse the problem exclusively from the point of view of exchange entitlement failure and with reference to the events in 1943 or thereabout. In other words, the crisis needs to be viewed from a broader and larger perspective.

The famine of 1943 and the famine-induced epidemic diseases that followed cost the Bengal province some three million lives.¹ A vast number of those who survived suffered decline in economic position and social status because they had to sell their land, plough cattle, tin sheds, utensils, tools, implements and ornaments to buy food. There was a large increase in the number of destitute and beggars.² Judged by any standard the famine was a great calamity for the province. The question what led to the crisis of such a magnitude naturally

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¹ Paul R. Greenough, *Prosperity and Misery in Modern Bengal: The Famine of 1943–44* (New York, 1982), p. 303.

² P.C. Mahalanobis, 'A Sample Survey of the After-Effects of the Bengal Famine of 1943,' *Sankhya: Indian Journal of Statistics*, Vol. 7, part 4 (1946).

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occasioned a great deal of discussion during the famine and the years thereafter. Recent years have witnessed a renewed scholarly interest in the subject and the impetus has been provided by Amartya Sen's essay on the Bengal famine published in the late 1970s and early 80s.³ Some commentators have described his approach as 'revolutionary'⁴ and 'path-breaking'.⁵ On the other hand, some scholars have taken a negative view. One critic has gone to the extent of saying, "What Professor Sen has said about famine, was known to my grandmother".⁶ This essay is an attempt to review the different opinions on the causes of the famine (popularly known as *panchasher manantar*), particularly Amartya Sen's proposition that there was no food availability decline (FAD). Secondly, some general comments are made on Amartya Sen's approach.

I

The contemporary non-official observers were almost unanimously of the view that the famine was 'man-made', no natural factors such as drought or flood caused such a shortage of rice as to make large-scale starvation and death inevitable. As the leading Calcutta daily the *Statesman*⁷ put it, "This sickening catastrophe is man-made. We say with deliberation that the present Bengal famine constitutes the worst and most reprehensible administrative breakdown in India since the political disorders of 1930–31". K.C. Neogy took the opinion that "this famine is primarily a state industry, and to my mind, in certain of its aspects it bears the hallmark of a genuine British manufacture".⁸ According to this view the famine was due to the negligence of the government, gross administrative inefficiency and implementation of wrong policies. Black-marketing and profiteering of a section of the traders, it was conceded, played a certain part in causing

³ Amartya Sen, "Starvation and Exchange Entitlements: A General Approach and Its application to the Great Bengal Famine", *Cambridge Journal of Economics* 1 (1977); *Poverty and Famines: An Essay on Entitlement and Deprivation* (New Delhi, 1994).

⁴ Sugata Bose, "Starvation amidst Plenty: The Making of Famine in Bengal, Honan and Tonkin, 1942–45", *Modern Asian Studies*, 24, 4 (1990).

⁵ Quadrat-I-Ilahi, "Amartya Sen's Famine Philosophy", *The Daily Star*, May 12, 2005.

⁶ *Ibid.*

⁷ *The Statesmen*, 23 September, 1943 cited by Omkar Goswami, "The Bengal Famine of 1943: re-examining the Data", *Indian Economic and Social History Review*, 27, 4 (1990).

⁸ Cited by K.C. Ghosh, *Famines in Bengal, 1770–1943* (Calcutta, 1945), p. 171.

hunger, suffering and death, but the main focus was on administrative inefficiency and negligence. The Famine Inquiry Commission which was appointed by the Government of India in 1944 to investigate the causes of the famine and to recommend steps for its prevention, took a different view. According to this Commission (henceforth FIC) the famine was caused by two factors. Firstly, during 1943 there was a 'serious shortage' in the total supply of rice available for consumption in Bengal as compared to the supply normally available. This was due to the shortage in (a) winter rice harvested towards the end of 1942 and early in 1943, (b) stock of old rice carried over from 1942, and (c) loss of import from Burma. Current supply of the three varieties of autumn, winter and summer rice was sufficient for 43 weeks' requirements and the carry-over from 1943 was sufficient for another six weeks. Thus, the total supply of rice in 1943 could have met 49 weeks' requirements of the province, leaving a deficit of about five per cent.⁹ Secondly, there was a phenomenal increase in price level. Consequently food grains prices went beyond the purchasing power of those who normally depended on the market supply for part or the whole year and there was large-scale starvation and death.

In a dissenting note M. Afzal Hussain, a member of the FIC, argued that "during recent years, and particularly in the beginning of 1943, there was no carry-over". For many years prior to 1943 there was no increase in rice production, but the population size was growing. Therefore, it is not valid to assume that there was any carry-over of rice in 1943. This meant that, according to Mr. Hussain, if we are to make any idea about the supply position in 1943 it must be done on the basis of the production level of only 1943. The *aman* crop available for consumption during January–September of this year was 5.5 million ton, but the actual requirement was of the order of 7.2 million tons.¹⁰ There was thus a deficit of about 1.5 million tons, or about 20 per cent. B.M. Bhatia seems to proceed on the estimates of Afzal Hussain.¹¹ George Blyn goes a step further. He argues that in 1942–43, cyclones and floods reduced the Bengal rice crop by about a third; this, coupled with the absence of imports from Japanese controlled Burma and inadequate relief, led to famines, epidemics

⁹ *Report of the Famine Inquiry Commission, Bengal* (Calcutta, 1945), p. 77. Henceforth also referred to as FIC or Commission.

¹⁰ *Ibid.*, pp. 179–199.

¹¹ B.M. Bhatia, *Famines in India: a Study of Some Aspects of the Economic History of India, 1860–1945* (Bombay, 1963), pp. 322–23.

aggravated by widespread starvation.¹² Paul Greenough¹³ estimates that even after deducting the losses due to the cut-off of Burmese imports, Midnapur Cyclone, flood and serious crop disease, 90 per cent of the usual supply of rice was available in 1943. Dipak Basu¹⁴ also takes the view that there was a certain decline in food supply in this year. But none of them takes the view that FAD as such was the primary factor behind the famine.

Amartya Sen rejects what he calls “the standard explanation of the famine”, that is, the view that the crisis was caused by food availability decline (FAD). He concedes that “1943 was not a very good year in terms of crop availability”, but “it was not by any means a disastrous year either”. He continues that this contention is supported by the statistics provided by the FIC itself. For these indicate that “the current supply for 1943 was only about five per cent lower than the average of the preceding five years. It was, in fact, 13 per cent higher than in 1941, and there was of course no famine in 1941”.¹⁵ Thus, Amartya Sen’s view that FAD did not cause the famine is based on two evidences: (a) the supply of food grains in 1943 was not too bad, and (b) 1941 was much worse in terms of food availability (Appendix 1), but there was no famine in this year. Famine occurred in 1943 because for a large section of population there was a marked decline in exchange entitlement to rice. Firstly, there was a dramatic shift in the ratio between agricultural wages and the price of rice between these two years. Consequently, by 1943 wage level of agricultural labourers as share of the price of rice declined by two-thirds i.e., the index of exchange rate stood at 34 (Appendix 3). Therefore, it is no wonder that this occupational category contributed most, in absolute terms, to the total number of deaths and destitution. Secondly, the study also shows the decline in exchange entitlement to rice of producers of other commodities such as mustard oil, cloth, bamboo, umbrella, milk, fish and hair-cuts.¹⁶ Consequently, there was large-scale starvation and death also among these groups of people. Amartya Sen undertakes the study to provide empirical support to his general proposition that famine in the sense of starvation of particular groups of people can

¹² Cited by Amartya Sen, *Poverty and Famines*.

¹³ Greenough, *Prosperity and Misery in Modern Bengal*, p. 99.

¹⁴ Dipak R. Basu, “Sen’s Analysis of Famine: A Critic” *Journal of Development Studies*, 22 (1986).

¹⁵ Amartya Sen, *Poverty and Famines*, p. 58.

¹⁶ *Ibid*, pp. 63–70.

occur without an overall FAD. As he puts it, "The FAD approach gives little clue to the causal mechanism of starvation, since it does not go into the relationship of people to food. Whatever may be the oracular power of the FAD view, it is certainly Delphic in its reticence".¹⁷ Sugata Bose has gone a step further. He does not make any attempt to estimate the level of food production in per capita terms, but suggests that the famine struck Bengal "amidst plenty".¹⁸ Thus while some commentators argued that there was a significant decline in food production, others, in particular Amartya Sen, have rejected this view.

II

Clearly the validity of Amartya Sen's proposition about FAD hinges on the relative level of food production in 1941 and 1943 compared to other preceding years. As pointed out by Peter Bowbrick, if his figures (and analysis) are proved wrong his whole argument will collapse.¹⁹ What is then needed is a brief review of the nature of the crop statistics and the manner in which these are used by Amartya Sen. Incidentally he draws his database from the FIC report and the FIC received the core statistics from the Agricultural Department. Now, it has been generally assumed that the crop statistics of all the parts of India in general and those of Bengal in particular are unreliable. The Royal Commission on Agriculture in India observed: "These (Bengal crop statistics) are merely guesses and not infrequently manifestly absurd guesses".²⁰ As the revenue demand in Bengal was permanently fixed, the government did not maintain, as it did in other provinces, any regular staff at the village level for reporting crop statistics, i.e. statistics on area under cultivation and yield per acre. In the absence of such a staff, crop acreage was reported by Khashmahal Tahsildar/Thana Officer/Circle Officer/District Agricultural Officer. It has generally been believed that area statistics reported by the government were grossly underestimated. Comparison with data available from independent sources confirms that this was generally

¹⁷ *Ibid*, p. 54.

¹⁸ Sugata Bose, "Starvation amidst Plenty".

¹⁹ Peter Bowbrick, "The Causes of Famine: A Refutation of Professor Sen's Theory", *Food Policy* 11, no. 2.

²⁰ *Report of the Royal Commission on Agriculture in India*, Vol.1 (London, 1928), p. 605. For further details on the nature of crop statistics see M. M. Islam, *Bengal Agriculture 1920-46: A Quantitative Study* (Cambridge, 1978), Ch. 1.

the case. Two independent sources deserve particular mentioning in this respect. These are: (a) the *District Survey and Settlement Reports* and (b) the crop statistics collected in 1945 through plot-wise survey.²¹ The statistics from the *Settlement Reports* used by the FIC for the purpose of comparison indicate that the range of underestimation was 20 per cent. On the other hand, a careful scrutiny of those available from the Ishaque Report suggest that the range was even higher (35 per cent). With regard to yield rate, crop-cutting experiments were conducted to estimate Standard Yield, and yield per acre of the major crops (except jute) in every year was expressed as percentage (Condition Factor) of the standard. Standard Yield was revised every five years. The experiments were not conducted in randomly selected fields and in any case the number of experiments was too small. What is more, Condition Factor was worked out purely on subjective considerations. In other words, these were “eye-estimates” without any scientific basis. Anyway, a comparison with data available from independent sources (i.e. data based on surveys conducted by the Indian Statistical Institute) indicate that the Agricultural Department overestimated the yield rate of *aman* and *aus* rice. In other words while the Agricultural Departments underestimated the crop acreage, it overestimated the yield rates. Therefore, it will be extremely difficult to be confident about the reliability of the physical volume of production of these crops at any particular point time.

In order to remove these elements of unreliability the FIC adjusted upward the *aman* acreage from 1927/28 onwards by 20 per cent and proposed the reduction the yield rates of *aman* by 1/15th and *aus* by 1/12th. The FIC was confident that the adjusted figures on rice output were “nearer the truth”.²² But at least two circumstances suggest that the FIC should not have been so confident. Firstly, as elaborated later, there are certain anomalies in the acreage data which are not taken care of by the FIC. Secondly, in arriving at the figures on *aus* and *aman* production the FIC did not revise the yield rates as planned. Thus, the estimates of the FIC about the physical volume of production cannot be taken as reliable either.

Still Amartya Sen builds up his thesis on the basis of the FIC’s database. Since he is primarily concerned with the *relative level* of rice production, especially in 1941 and 1943, he can indeed proceed on

²¹ *Government of Bengal, Agricultural Statistics by Plot to Plot Enumeration in Bengal 1944–45*, 3 parts (Calcutta, 1946). This is popularly known as Ishaque Report.

²² FIC, p. 208.

the basis of FIC's data. But in that case one condition needs to be met: the margin of error in yield rates and area statistics should be shown to have been more or less the same over the years. A close look at the relevant statistical series suggests that in the case of *aman* acreage the margin can be assumed to have been more or less uniform up to 1940/41, not for the subsequent two years (1941/42 and 1942/43). This is because in 1941/42 the Agricultural Department revised upwards its estimates,²³ thus making allowance for its underestimation in the earlier years. Thus, whereas the annual average of *aman* acreage during the preceding four years was 15715 thousand acres (Table 1), in 1941/42 it was about 10 per cent higher. The implication is obvious, the acreage data prior to 1941/42 was underestimated by 30 per cent, not by 20 per cent as assumed by the FIC. The FIC proceeded on the assumption that *aus* (and *boro*) acreage can be taken as reliable. But even a cursory look at the statistics makes it abundantly clear that prior to 1940/41 *aus* area was underestimated by more than 10 per cent. Therefore area under *aman* and *aus* rice prior to 1941/42 should be revised upward by 30 and 10 per cent respectively. With regard to yield statistics, it will be extremely difficult to guess whether the margin of error was uniform. Incidentally yield rates depend on a larger number of factors than does area under cultivation. Thus it will be risky to attempt some kind of adjustment in yield statistics. This means that if an attempt has to be made to estimate rice production, the official yield statistics have to be taken at their face value. Incidentally, this is exactly what the FIC did.

Since Amartya Sen exclusively depends on the database of the FIC he is not supposed to be aware of these anomalies in the acreage statistics. The same is true with regard to the FIC, for it received the data from the Director of Agriculture. However, it is clear that some kind of further adjustment, especially in the case of *aman* production will be in order. Accordingly *aman* acreage (that is production) for the period up to 1940/41 is revised by 30 per cent. For the next two years the adjusted figures of FIC are accepted. This means an improvement in the relative position of 1941. There is yet another consideration for which *aman* production for this year should be said to have been higher than what was recorded by the FIC (4315 thousand tons). For calculation on the basis of data available from the *Season and Crop*

²³ M. M. Islam, *Bengal Agriculture*, p. 30.

TABLE 1
Statistics on Acreage and Production

Years	<i>Aman</i> acreage			<i>Aman</i> production			<i>Aus</i> acreage	<i>Aus</i> production	Rice and wheat current supply		Index of total food grains	
	Official	Adjusted	Revised	Official	Adjusted	Revised	Official	Official	Adjusted	Revised	Adjusted	Revised
1937/38	15923	19107	20700	6873	8247	8935	5727	1473	102107	10905	123	115
1938/39	15836	19003	20587	5963	7155	7752	5742	1846	9787	10384	118	110
1939/40	16096	19315	20925	6504	7805	8455	5416	1601	10196	10846	122	115
1940/41	14917	17900	19392	4315	5178	6297	6485	2250	8332	9451	100	100
Average	15693*						5688**	—				
1941/42	16914	20297	29297	7396	8876	8876	6507	1694	10947	10947	131	116
1942/43	16207	19449	19449	5020	6024	6024	6500	1865	9235	9235	110	98
Three years' moving average												
1938-40											121	113
1939-41											113	108
1940-42											118	110
1941-43											114	105

Notes: (a) area in 000 acres, (b) production in 000 tons.

(c) *stands for average of four years (1937/38-1940/41).

(d) **stand for average of three years (1937/38-1939/40).

(e) Adjusted figures are those of the FIC used by Amartya Sen, *Poverty and Famines: Essay on Entitlement and Deprivation* (New Delhi, 1984), Appendix 1.

(f) Revised figures are those estimated in this essay.

Report indicates that the level of *aman* production in this year was 4844 thousand tons.²⁴ A multiplication of this figure by 1.30 gives a production level of 6055 thousand tons. As mentioned earlier, *aus* production for the period prior to 1941/42 should also be revised upward by at least 10 per cent. But this is not being done to make allowance for any overestimation in *aman* production which may have been caused by its upward revision by 30 per cent. Finally, calculation on the basis of data available from the *Season and Crop Report* shows that *aus* production in 1943 was 3023 thousand tons. But the FIC used the figure 2390 thousand tons perhaps on the assumption that the Agricultural Department over reported the level of production by about 20 per cent. In all probability this will be a valid assumption, because there is no ground to justify the marked expansion of *aus* acreage in 1943/44. Therefore, no allowance seems to be in order on this count.

Production statistics for different years arrived at on the basis of such a procedure are presented in Table 1. From these statistics it appears that the level of food grains supply was a little higher in 1941 than in 1943. Thus the negative part of Amartya Sen's arguments (that there was no famine in 1941 even though supply was lower) is not borne out by the alternative set of data generated in this essay. Secondly, *aman* production in 1943 was 32 per cent lower than in 1942, but total food grain supply in the former year was only 15 per cent lower. Thirdly, on the basis of three years' moving average food grains availability seems to have been five per cent lower in 1943. Finally, compared to the average of the preceding quinquennium the decline was of the order of 13 per cent. Thus, clearly FAD in 1943, whether estimated on the basis of moving or quinquennial average or annual statistics, was not of such a magnitude that it could have caused a famine of such proportions. Only a moderate increase in the price level could have been justified even if the decline was of the order of 15 per cent.

Thus, one of the two contentions behind Amartya Sen's proposition about Bengal famine (i.e. there was no significant reduction in food

²⁴ Crop statistics were first published in the *Season and Crop Reports*. The final forecast figures were published in the *Agricultural Statistics of Bengal*. Sometimes these statistics were revised, but not significantly. This is clear from the fact that in all other years there are only marginal differences between production estimates based on *Season and Crop Report* data and those used by the FIC. Therefore, it will be reasonable to assume that there is a genuine error in the data on *aman* production reported by the FIC for 1940–41.

availability in 1943) seems to be borne out by the alternative estimates presented in this essay. The other evidence—food availability was even lower in 1941, but there was no famine in that year—does not appear to be tenable. Still his analysis may be regarded as “revolutionary” or “path-breaking”. For historically speaking, most famines were preceded by crop failure due to some natural factors and therefore it has seemed obvious to many commentators that famines are usually caused by food availability decline or shortage.²⁵ According to Amartya Sen the Bengal famine differed from this pattern in the sense that it was independent of a dramatic FAD: there was a crisis because a whole range of ‘man-made’ factors (as well as and ‘indifferent’ *aman* harvest in 1943) led foodgrain prices increase much faster than wage rates and the prices of certain goods and services in the rural areas. These ‘man-made’ factors were wartime inflation leading to increased demand, speculative withdrawal, panic purchase, boat and rice “denial” policy, prohibition of exports from other provinces. In other words the famine was primarily due to the extraordinary situation created by the war and the acts of omission and commission by the government as also the response of the buyers, sellers and traders in the face of this situation.

However, on close scrutiny it appears that this was basically the position also of the FIC which Amartya Sen takes to task for advocating the FAD view. For though the FIC admits that the total supply situation was worse in 1943 it hastened to add that the deficit was not however “such as to make starvation on a wide scale inevitable, provided that the trade was capable of maintaining the distributions of available supplies subject only to a moderate rise in the price level”.²⁶ At another point the FIC is more categorical,

The remarkable feature of the Bengal famine was that the rise in the price of rice was one of the principal *causes* of the famine. This, as far as we are aware, makes it *unique in the history of famine in India* (emphasis added). The great majority in Indian famines have been caused by drought and widespread failure of crops over wide areas. Floods, hail, and cyclones have on rare occasions produced the same effect. War, and civil disorder played a part in some of the great famines of the 18th century and earlier eras. In large famines produced by such calamities the shortage of grain was naturally

²⁵ W.R. Aykroyd, *The Conquest of Famine* (London, 1974). For this view about the famines that struck greater Bengal in the second half of the 19th century see Malabika Chakrabarti, *The Famine of 1896–97 in Bengal: Availability or Entitlement Crisis?* (New Delhi, 2004).

²⁶ FIC, p. 15.

reflected in a rise in price, but the latter was a secondary phenomena, and not a primary cause of the famine.²⁷

In this connection the Commission also pointed out that the Famine Code which took shape during the last quarter of the nineteenth century was evolved as a measure against typical 'drought famines'. It was then obvious, the Commission further argued, that the Bengal famine 'as it developed by the middle of 1943 and the kind of famine with which the Code is concerned were two very different things'. Thus, clearly the FIC was not actually advocating the FAD view. On the other hand, anticipating Amartya Sen's proposition, it took the view that price-rise of food grains (grossly disproportionate with the supply position) basically lay at the root of the crisis. Incidentally, several contemporary and latter day commentators argued in the same vein. For example, Aykroyd, who was a member of the FIC, subsequently said, "sheer lack of rice was a less important cause of starvation and death than its enormous price".²⁸ Again, Bhatia suggested that the famine of 1907–8 was the last drought related famine in India.²⁹ The Bengal Provincial Congress Committee observed that the "famine was more man-made than an act of God".³⁰

Why does then Amartya Sen believe that the FIC advocated the FAD view? In a way the Commission itself is responsible for this misunderstanding. For though it argued that the deficit was of the order of only five per cent (the available supply being sufficient for 49 weeks' requirements) and it put in considerable effort to explain why there was more than a proportionate rise in the price level and consequently food grains went beyond the purchasing power of a vast section of the population, at the same time the FIC also pointed out that there was a "serious shortage"³¹ of food grains and this was "one of the basic causes"³² and the rise in prices was the "second basic cause"³³ of the famine. If by its own admission the Bengal famine was unique in the history of Indian famines in the sense that the price rise in the Calcutta market was 100 per cent between March and May (1943) even though the deficit in food grains supplies was only five per cent,

²⁷ *Ibid*, pp. 96–97.

²⁸ Aykroyd, *The Conquest of Famine*, p. 74.

²⁹ Bhatia, *Famine in India*, 2nd edition (1967), p. 270.

³⁰ I concurred with this opinion in my *Bengal Agriculture*, pp. 59–60. The comment by the Congress Committee is cited on p. 60.

³¹ FIC, p. 77.

³² *Ibid*, p. 103.

³³ *Ibid*, p. 104.

price rise cannot be treated as the “second basic” cause: it should be described (as the FIC did earlier) as the primary factor behind the famine. Thus considered it is clear that there is hardly any difference between the position of FIC and Amartya Sen.³⁴

Before proceeding further with Amartya Sen some comments on Peter Bowbrick’s analysis will be in order. According to him, crop statistics of Bengal were not only unreliable, but also wrong.³⁵ Therefore, he is in favor of rejecting them and instead depending on the qualitative evidence supplied by trade circles which were talking of the worst *aman* crop in 1943 in 20 years. Several aspects of Peter Bowbrick’s contention may be noted. Firstly, the defects of the available crop statistics are well-known, but this cannot be any justification for relying on the qualitative assessment of certain trade circles only. If such a statement has at all to be taken seriously one should see what the reaction of contemporary observers was in 1941. For Amaratya Sen’s proposition about famine without any drastic FAD in 1943 is partly based on the supply situation in 1941 as well. As a matter of fact Peter Bowbrick does not at all address the question: if the decline in *aman* harvest was a cause of the famine in 1943 why a larger decline in 1941 was not followed by a similar crisis. Secondly, if according to Peter Bowbrick these statistics were wrong and therefore not at all useable he should not base any of his argument on them. But he uses these data in Table 1. Thirdly, it is extremely difficult to understand why he thinks that the margin of error in Amartya Sen’s estimate on the level of production could be of the order of 3000 per cent. Finally, in an attempt to ‘destroy’ his argument about FAD in 1943 Peter Bowbrick ‘arbitrarily’ assumes that there was a “carry over” of 8.5 million tons in January 1939. His decision to proceed from an arbitrary assumption about food surplus and that too at the level of about one year’s *aman* production is even more incomprehensible.

³⁴ However, in one respect there is a significant difference. The FIC did not confine its attention to the failure of the government in averting the crisis, it discussed at considerable length government’s failure to mitigate the sufferings of the people as the crisis unfolded (FIC, Chapters VII–IX). For a proper understanding of the severity of the famine failure of the government in the latter field needs to be taken into consideration more fully than Amartya Sen does.

³⁵ Peter Bowbrick, “The Causes of Famine”. Several other scholars have taken part in the debate. But it is beyond the scope of this paper to comment on all or most of them.

III

Several other aspects in Amartya Sen's analysis deserve to be pointed out. Firstly, the scale of devastation cannot be understood by considering the events of the years immediately preceding the famine. A realistic assessment of the famine should begin earlier, at least from the turn of the 1920s. In this respect Paul Greenough, Omkar Goshwami and Utsa Patnaik's work make a significant contribution. Several surveys on per capita daily³⁶ cereal consumption by various economic groups of Bengal were made between 1936 and 1942. Their findings indicated that most rural Bengalis consumed something between 397 and 462 grams of cereals per capita daily. Does it reflect a secular decline in food grains availability over the past years in per capita terms? Given the nature of crop statistics it will be difficult to guess as to what was actually happening. But several considerations suggest that there was a long-term decline. (a) From about the turn of the twentieth century food production in Bengal had remained stagnant. This was due to the fact that cultivation had virtually reached its natural limits and yield rates had hardly improved.³⁷ On the other hand, population size had grown from 42.9 million in 1901 to 61.6 million in 1941.³⁸ (b) Since 1929 Bengal had become a net importer of foodgrains, but imports constituted a small part of the total production. Thus considered, it is quite possible that average consumption levels declined over time. What is, however, more important to note in this connection is the implication of average consumption: while certain groups in the population were eating more than 482 grams around 1940, others were necessarily consuming less.³⁹ In other words, even if it is accepted that there was no secular decline in food availability at the aggregate level, long before 1943 a significant section of the rural people was experiencing FAD and living at the margin of subsistence, or in the words of a senior British official, in "semi-starved" condition.⁴⁰ This section included

³⁶ Greenough, *Prosperity and Misery*, pp. 71–81.

³⁷ Islam, *Bengal Agriculture*, Chapter 2.

³⁸ Greenough, *Prosperity and Misery*, p. 62.

³⁹ *Ibid*, p. 80.

⁴⁰ In a confidential note by L.G. Pinnel, Director of Civil Supply, Government of Bengal, pointed out, "It may well be asked why in the circumstances... a famine did not take place earlier? The reply to this is that though there was no actual "famine" as it is usually understood, quite a large section of the people were habitually living in a semi-starved condition for many years". The same view is taken by

1.22 million sharecropping and 2.25 million wage-labour households as also barber, blacksmiths, carpenters, the poor weavers, destitutes and beggars.⁴¹ Incidentally one of the slogans raised by Fazlul Huq's Krishak Praja Party during its campaigns in the 1937 Bengal Assembly election was 'dal-bhat,' or plain rice (the staple food of the Bengalis) and lentil-like pulse. As pointed out by Paul Greenough, such a slogan was clearly indicative of the fact that in the 1930s millions of Bengalis were failing to get sufficient rice to eat.⁴²

An important factor behind the economic scenario was the depression experienced by India from 1929 onwards. Agricultural prices started falling worldwide earlier than industrial prices and they fell further than did manufacturing goods prices. Within India itself the industry-agriculture terms of trade turned sharply against agriculture. The income terms of trade (measuring the purchasing power in sector against the other had declined for agriculture from a world wary high by 33 per cent up to 1929–30. This was now followed by a further decline of 17 per cent up to 1934/35. The impact of this development on the rural population was very severe, especially because their cash needs did not decline proportionately. While the landlords and rich peasants faced a squeeze on their surplus the middle peasants had to supplement their falling income from cultivation by working as wage laborers and the small ones faced pauperisation of the kind faced by the poor and landless peasants. Incidentally there was a considerable volume of "distress" sale of land held by a section of the rural population. The impact of the depression was exacerbated by the colonial government's strict adherence to balanced budget and the consequent cuts in public spending on irrigation works, roads and railways. At a time when large masses of small and poor peasants were forced to join the labour market the employment opportunities were being reduced and wage rates lowered.⁴³ According to the Bengal provincial Kishan Shabha wage rate was so low that "even if the laborer is so lucky as to secure 30 days of work in a month . . . he will not earn enough to enjoy as much as a convict in a jail".⁴⁴ Overall, the impact

M. Alamgir, *Famine in South Asia: Political Economy of Mass Starvation* (Cambridge, 1980), p. 79.

⁴¹ Greenough, *Prosperity and Misery*, p. 66.

⁴² *Ibid.*, p. 70.

⁴³ This part of the discussion is based on Utsa Patnaik, *The Long Transition: Essays in Political Economy* (New Delhi, 1999), pp. 331–34.

⁴⁴ Cited by Boudhayan Chattapadhyaya, "Notes Towards an Understanding of the Bengal Famine of 1943". *CRESSIDA Transactions* Vol. 1 (Summer 1981), No. 1.

of the depression was to enlarge the group of asset-less and asset-poor people in rural areas.

The implication of the fact that a large section of the population was asset-less and living at the margin of subsistence is obvious and very important: it did not make famine inevitable, but the balance between actual starvation and bare subsistence was so delicate “that the slightest tilting of the scale in the value and supply of food was enough to put it out of the reach of many and to bring large classes within the range of famine”.⁴⁵ This was Bengal’s predicament in 1943. As further explained by the FIC

It is . . . necessary to draw attention at the outset to the condition of the people in normal times. The standard of living was in general low. Population was growing rapidly, leading to increased pressure on available land suitable for cultivation . . . At the best of times . . . a section of the poorer classes . . . did not get enough to eat and their diet, largely composed of rice, was of poor nutritive quality . . . Dispensary doctors in Bengal reported that only 22 per cent of the population was well nourished while 31 per cent were badly nourished . . . It is difficult to accept these figures as showing conclusively that nutritional conditions in Bengal were worse than elsewhere in India. Nevertheless, the results of these investigations may be quoted as indicating in a general way the unsatisfactory state of nutrition of the people of rural Bengal 10 years previous to the famine. Poverty and malnutrition left a section of the population with few reserves . . . For them there was no margin of safety and little probability of tightening the belt. The fact that such conditions are common to most other provinces of India which escaped the famine, does not detract from their fundamental importance. They provide a background against which the events which led to the widespread starvation in 1943 must be viewed.⁴⁶

Amartya Sen overlooks these aspects of the problem. In the theoretical part of his discussion he points out that poverty does not necessarily mean starvation. He admits that “Starvation is a normal feature in many parts of the world”, but he cautions that in

analysing starvation in general it is important to make clear distinctions between three different issues: (1) lowness of the typical level of food consumption; (2) declining trend of food consumption; and (3) sudden collapse of the level of food consumption. Famine is chiefly a problem of the third kind, and while it can—obviously—be helped by the first two features, it often does not work that way.⁴⁷

⁴⁵ FIC p. 16.

⁴⁶ FIC pp. 6–7.

⁴⁷ Amartya Sen, *Poverty and Famines* pp. 40–41.

Thus, though he admits that generally there is a link between famines and the low level of food consumption and its secular decline, he argues that there was no such relationship in the case of the Bengal famine of 1943. But it is suggested that for a proper understanding of the intensity of the crisis allowance needs to be made for this long-term decline in food consumption of a large section of the rural community as this should have been slowly but surely sapping their strength and vitality.⁴⁸

Secondly, Amartya Sen discusses how the exchange entitlement of wage-labourers and certain other groups declined or the price of rice went out of their purchasing power and consequently they became the victims of the famine. But he does not mention the fate of the small and marginal farmers (including *bargadars*) who partly depended on the market supply but had not joined the rank of agricultural labour. They depended partly on exchange and partly on direct entitlement. In other words, their subsistence needs were met partly from their own production and partly from market supply. Incidentally, in the late 1930s, 36 per cent of the rural families held less than two acres per holding.⁴⁹ In East Bengal districts many of them had taken to jute cultivation to meet part of their subsistence needs. The idea was that since jute cultivation fetched gross higher revenue per acre the earnings from jute could be used to buy a larger quantum of rice for domestic consumption.⁵⁰ Thus, in the case of these farmers the relative decline in the price of jute needs to be indicated. Incidentally 10 of the 15 'very severely affected' subdivisions were in the jute growing heartland, as were six of the 'severely affected areas'.⁵¹ Thirdly, in order to avoid starvation a section among them dispensed of their non-labour assets—land, plough cattle, tin sheds, ornaments etc. But the price of rice must have increased more than the prices of these assets. Consequently, the sale of non-labour assets failed to prevent distress to these people. Once again, Amartya Sen refers to these possibilities

⁴⁸ Greenough has argued that the major cause of the epidemic cholera, smallpox and malaria was the nutritionally weakened state of the rural population and the unusual concentration of migrant destitute. For details see his *Prosperity and Misery*, p. 163.

⁴⁹ Government of Bengal, *Report of the Land Revenue Commission, Bengal*, vol. 2 (Calcutta, 1940), Table VIII(A).

⁵⁰ Sugata Bose, *Agrarian Bengal: Economy, Social Structure and Politics* (Cambridge, 1986), pp. 46–58.

⁵¹ Omkar Goswami, *Industry, Trade and Peasant Society*, p. 187.

in the theoretical part of his discussion,⁵² but he does not make any reference to them in his discussion on the famine itself. Fourthly, according to Amartya Sen the war efforts created an inflationary spiral by creating access demands for foods. According to him, "... if we take a group of people who are under-fed and increase the income of half of them, the other half would be pushed towards starvation because the demand created by the better half would exhaust the possible supply". But as Dipak Basu points out this contention of people with 'part-filled' bellies creating market disruption because of sudden increase of their financial means does not seem realistic.⁵³ Fifthly, the shortfall in *aman* production in 1943 was not as 'moderate' or 'indifferent' as he would have us believe: the decline was a marked one (30 per cent compared to 1942 and 20 per cent compared to the previous quinquennium). This decline in *aman* production made a more significant contribution to the 'exceptional shortfall in market release' (and the consequent sharp rise in food grain prices) than he seems to concede.

Finally, as pointed out by Paul Greenough,⁵⁴ Amartya Sen's discussion of the famine is one-sided because of his assumption that the sole source of food for all sections of the rural population must have been the market. The actual situation is more complex. (a) As the food prices registered a sharp rise the patron-client ties came under severe strain in the sense that the dominant section of the rural population disengaged themselves from their fixed obligations to feed their dependants. The question of exchange entitlement is not relevant also in the case of such dependants. This is true also in the case of the larger section of the destitute who normally depended on non-market sources (begging and charity) for their subsistence. Incidentally, at the beginning of 1943 there were 7.5 lakh destitute in the province and the death rate among them should have been very high, perhaps one of the highest. Again, this is the group which must have suffered in large number from the outbreak of famine-induced epidemic diseases like cholera, influenza and malaria. (b) A section of the small farmers must have been depending on loans in cash and kind to meet the deficit in their consumption needs. Indeed, the fact that the small farmers were resorting to distress borrowings at high rates of interest is too well-known. But in the given situation in 1943 such loans must have become hard to come by. Even if cash loans were

⁵² Amartya Sen, *Poverty and Famines*, p. 4.

⁵³ Dipak R. Basu, "Sen's Analysis of Famine: A Critic".

⁵⁴ Greenough, *Prosperity and Misery*, p. 70.

available at previous levels these were not adequate to purchase the desired quantum of rice. Amartya Sen does not pay any attention to this aspect of the problem.

To conclude, theoretically speaking, there is little scope for quarrel with the proposition that famine in the sense of starvation of particular groups of people can occur without an overall decline in food availability. Given the nature of the available crop statistics it will be risky to take a firm position with regard to the question whether this was true in the case of the Bengal famine. But a closer look at these statistics seems to lend support to such a proposition. In other words, Amartya Sen's argument that the Bengal famine was not caused by any dramatic decline in total food supply in 1943 seems to be a valid one. However, the negative side of his contention (food supply in 1941 was lower) does not seem to be tenable. Secondly, a careful reading of the report of the FIC indicates that basically there is no difference between its and Amartya Sen's assessment about the primary cause behind the famine. For, by coincidence or otherwise, according to him and FIC there was a FAD to the extent of only five per cent, but due to a whole range of other circumstances this was followed by a grossly disproportionate rise in the food grain prices leading to large-scale starvation and death. Several other contemporary observers also took the view that the crisis was primarily due to 'man-made' factors. Thirdly, the crisis unfolded against a background in which a large section of the rural people were suffering from poverty and malnutrition. This did not cause the famine, but it aggravated the impact of the crisis when it struck. Amartya Sen completely overlooks this aspect of the problem. Fourthly, the decline in exchange entitlement was due not only to the fact that the wage level of agricultural labourers and prices of goods and services sold by fishermen, transport workers, paddy huskers, craftsmen and non-agricultural labourers did not increase proportionately with the increase in the price of rice. It was also caused by the relative decline in the prices of jute and disposable assets of the poorer section of the rural people—lands, implements, tin sheds, utensils, plough cattle. Thus, Sen's definition of exchange entitlement is rather narrow. Finally, a section of the rural people perished because they had no purchasing power at all, they all lived on charity. Thus, overall it is probable that the crisis of 1943 was not a reflection of a serious FAD, but for a proper understanding of its severity the famine needs to be viewed from a larger and longer perspective.

APPENDIX 1
Statistics of Food Supply

Period 1	Official estimate of <i>aman</i> production 2	Adjusted <i>aman</i> production 3	Net imports of rice 4	Current supply of rice (official) 5	Adjusted total rice production 6	Adjusted current supply of rice 7	Rice & wheat adjusted current supply 8	Index of total food grains 9	Index of per capita food grains availability 10
<i>Annual data</i>									
1938	6,873	8,247	0.033	8,507	9,848	9,981	10,217	123	127
1939	5,963	7,155	0.382	8,304	9,114	9,596	9,787	118	120
1940	6,504	7,805	0.258	8,481	9,524	9,882	10,196	122	123
1941	4,315	5,178	0.223	6,991	7,631	7,954	8,332	100	100
1942	7,396	8,876	-0.102	9,194	10,776	10,774	10,947	131	130
1943	5,020	6,024	0.264	7,892	8,632	8,896	9,235	111	109
<i>2 years moving averages</i>									
1938-39				8,406		9,789	10,002	120	123
1939-40				8,393		9,739	9,992	120	122
1940-41				7,736		8,918	9,264	111	112
1941-42				8,093		9,364	9,640	116	115
1942-43				8,543		9,835	10,091	121	119
<i>3 years moving averages</i>									
1938-40				8,431		9,820	10,067	121	123
1939-41				7,925		9,144	9,438	113	114
1940-42				8,222		9,537	9,825	118	118
1941-43				8,026		9,208	9,505	114	113

Source: Amartya Sen, *Poverty and Famines: An Essay on Entitlement and Deprivation* (New Delhi, 1984), p. 61.

Note: Million tons for columns 2-8.

APPENDIX 2

Price of coarse rice in Calcutta, January to December 1943 (rupees/maund)

Date	Price	Date	Price
January 4	11/4	July 28	31/3
January 20	12/8	August 2	31/5
February 3	13/2	August 9	31/1
February 17	13/12	August 16	32/3
March 3	15/0	August 23	34/8
March 17	19/6	August 30	34/13
March 22	21/4	September 6	31/9
March 29	22/0	September 7	31/12
April 5	21/3	September 13	26/0
April 12	20/7	September 20	25 to 26/0
April 19	19/5	September 27	20/0
April 26	21/0	October 4	20/0
May 3	21/1	October 11	20/0
May 10	25/0	October 18	20/0
May 17	30/10	October 25	20/0
May 24	29/2	November 1	20/0
May 31	30/6	November 8	20 to 22/0
June 7	31/8	November 15	20 to 22/0
June 14	31/10	November 22	20 to 22/0
June 21	31/0	November 29	20 to 22/0
June 28	31/0	December 13	17/0
July 7	30/8	December 20	17/0
July 14	30/8	December 27	17/0
July 21	30/11		

Source: Paul R. Greenough, *Prosperity and Misery in Modern Bengal: The Famine of 1943-44* (New York, 1982), p. 115.

APPENDIX 3

Wage and Food Grains Price Index

Year	Wage index	Food grains price index	Index of exchange rate
1939-40 (1940)	100	100	100
1940-41 (1941)	110	109	101
1941-42 (1942)	115	160	72
1942-43 (1943)	125	385	32
First half of 1943-44 (1943)	130	385	34

Source: Amartya Sen, "Starvation and Exchange Entitlements: A General Approach and Its application to the Great Bengal Famine", *Cambridge Journal of Economics* 1 (1977).